

Integral Indefinida

- Integral é a operação matemática inversa da derivada.

DERIVADA ($f'(x)$)	INTEGRAL ($f(x)$)
$f'(x) = x^1$	$f(x) = \frac{x^2}{2}$ $\left(\frac{x^2}{2}\right)' = \frac{\cancel{2}x}{\cancel{2}} = x$
x^2	$\frac{x^3}{3}$ $\left(\frac{x^3}{3}\right)' = \frac{\cancel{3}x^2}{\cancel{3}}$
x^{10}	$\frac{x^{11}}{11}$
$x^N, \quad N \neq -1$	$\frac{x^{N+1}}{N+1}$
$f'(x) = 5 \cdot x^0$	$f(x) = 5x$ $(5x)' = 5$
10	$10x$ $(10x^1)' = 10x^0 = 10$
-2	$-2x$

$$K \in \mathbb{R}^*$$

$$O(\text{zero})$$

$$f'(x) = 5x$$

$$5x^2$$

$$5x^{10}$$

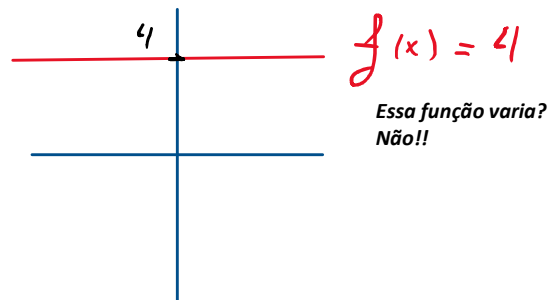
$$K f'(x)$$

$$\text{Sen } x$$

$$\cos x$$

$$Kx$$

$$K$$



$$f(x) = 5 \cdot \frac{x^2}{2} \quad \left(\frac{5x^2}{2} \right)' = \frac{5}{2} \cdot (2x) = 5x$$

$$\frac{5x^3}{3} \quad \left(\frac{5x^3}{3} \right)' = \frac{5}{3} \cdot \cancel{3} x^2$$

$$\frac{5x^{11}}{11}$$

$$K f(x)$$

$$- \cos x$$

$$\text{sen } x$$

$$f'(x) = 2^x$$

$$3^x$$

$$e^x$$

$$f(x) = \frac{2^x}{\ln 2} \quad \left(\frac{2^x}{\ln 2} \right)' = \frac{2^x \ln 2}{\ln 2}$$

$$3^x / \ln 3$$

$$e^x$$

$f(x) = x^n$ (polinomial): x na base

$f(x) = n^x$ (exponencial): x no expoente

Piada ruins: "festa das funções"

$$f(x) = n^x \rightarrow f'(x) = n^x (\ln n)$$

$$f(x) = 2^x \rightarrow f'(x) = 2^x (\ln 2)$$

$$f(x) = 3^x \rightarrow f'(x) = 3^x (\ln 3)$$

$$f(x) = e^x \rightarrow f'(x) = e^x (\ln e) = e^x$$

$= 1$

Notação para Integral:

$$\int f(x) dx = F(x) + C$$

$C \in \mathbb{R}$, a fim de indicar que há infinitas integrais para a função $f(x)$

Observação:

Para toda função há infinitas integrais.

Exemplo: $f'(x) = 2x$

$$\rightarrow f(x) = \cancel{2} \frac{x^2}{\cancel{2}} = x^2$$

E se derivarmos $g(x) = x^2 - 2$?

$$g'(x) = 2x$$

$$g'(x) = 2x$$

$$f'(x) = 2x \quad \begin{array}{l} \nearrow x^2 - 2 \\ \rightarrow x^2 + 1 \\ \searrow x^2 + 10 \end{array}$$

Derivada $2x$

Integral

EXERCÍCIOS

$$1) \int (4x + 7) dx = \frac{4x^2}{2} + 7x$$

$$= 2x^2 + 7x + C$$

$$2) \int (6x^5 - 8x^4 - 9x^3) dx = \frac{6x^6}{6} - \frac{8x^5}{5} - \frac{9x^4}{4}$$

$$= x^6 - \frac{8}{5}x^5 - \frac{9}{4}x^4 + C$$

$$3) \int (7x^{\frac{2}{5}} + 8x^{-\frac{4}{5}}) dx = 7 \cdot \frac{x^{\frac{2}{5}+1}}{\frac{2}{5}+1} + 8 \cdot \frac{x^{-\frac{4}{5}+1}}{-\frac{4}{5}+1}$$

$$= 7 \cdot \frac{x^{7/5}}{7/5} + 8 \cdot \frac{x^{1/5}}{1/5}$$

$$= + \cdot \frac{x}{\frac{7}{5}} + 8 \frac{x^{1/5}}{1/5}$$

$$= 5x^{7/5} + 40x^{1/5} + C$$

$$4) \int (\sin x - 2^x + 3e^x) dx$$

$$= -\cos x - \frac{2^x}{\ln 2} + 3e^x$$